CHAPTER 20

Volatility Smiles and Volatility Surfaces

How close are the market prices of options to those predicted by the Black–Scholes–Merton model? Do traders really use the Black–Scholes–Merton model when determining a price for an option? Are the probability distributions of asset prices really lognormal? This chapter answers these questions. It explains that traders do use the Black–Scholes–Merton model—but not in exactly the way that Black, Scholes, and Merton originally intended. This is because they allow the volatility used to price an option to depend on its strike price and time to maturity.

A plot of the implied volatility of an option with a certain life as a function of its strike price is known as a *volatility smile*. A three-dimensional plot of the implied volatility as a function of both strike price and time to maturity is known as a *volatility surface*. This chapter describes the volatility smiles and volatility surfaces that traders use in equity and foreign currency markets. It explains the relationship between a volatility smile and the risk-neutral probability distribution being assumed for the future asset price. It also discusses how traders use volatility surfaces as option-pricing tools.

20.1 IMPLIED VOLATILITIES OF CALLS AND PUTS

This section shows that the implied volatility of a European call option is the same as that of a European put option when they have the same strike price and time to maturity. This is a particularly convenient result. It shows that when talking about a volatility smile or volatility surface we do not have to worry about whether the options are calls or puts.

As explained in earlier chapters, put–call parity provides a relationship between the prices of European call and put options when they have the same strike price and time to maturity. With a dividend yield on the underlying asset of q, the relationship is

$$p + S_0e^{-qT} = c + Ke^{-rT} \tag{20.1}$$

As usual, c and p are the European call and put price. They have the same strike price, K, and time to maturity, T. The variable S_0 is the price of the underlying asset today, and r is the risk-free interest rate for maturity T.

A key feature of the put–call parity relationship is that it is based on a relatively simple no-arbitrage argument. It does not require any assumption about the probability distribution of the asset price in the future. It is true both when the asset price distribution is lognormal and when it is not lognormal.

Suppose that, for a particular value of the volatility, p_{BS} and c_{BS} are the values of European put and call options calculated using the Black–Scholes–Merton model. Suppose further that p_{mkt} and c_{mkt} are the market values of these options. Because put–call parity holds for the Black–Scholes–Merton model, we must have

$$p_{BS} + S_0e^{-qT} = c_{BS} + Ke^{-rT}$$

In the absence of arbitrage opportunities, put–call parity also holds for the market prices, so that

$$p_{mkt} + S_0e^{-qT} = c_{mkt} + Ke^{-rT}$$

Subtracting these two equations, we get

$$p_{BS} - p_{mkt} = c_{BS} - c_{mkt} \tag{20.2}$$

This shows that the dollar pricing error when the Black–Scholes–Merton model is used to price a European put option should be exactly the same as the dollar pricing error when it is used to price a European call option with the same strike price and time to maturity.

Suppose that the implied volatility of the put option is 22%. This means that $p_{BS} = p_{mkt}$ when a volatility of 22% is used in the Black–Scholes–Merton model. From equation (20.2), it follows that $c_{BS} = c_{mkt}$ when this volatility is used. The implied volatility of the call is, therefore, also 22%. This argument shows that the implied volatility of a European call option is always the same as the implied volatility of a European put option when the two have the same strike price and maturity date. To put this another way, for a given strike price and maturity, the correct volatility to use in conjunction with the Black–Scholes–Merton model to price a European call should always be the same as that used to price a European put. This means that the volatility smile (i.e., the relationship between implied volatility and strike price for a particular maturity) is the same for European calls and European puts. More generally, it means that the volatility surface (i.e., the implied volatility as a function of strike price and time to maturity) is the same for European calls and European puts. These results are also true to a good approximation for American options.

Example 20.1

The value of a foreign currency is \$0.60. The risk-free interest rate is 5% per annum in the United States and 10% per annum in the foreign country. The market price of a European call option on the foreign currency with a maturity of 1 year and a strike price of \$0.59 is 0.0236. DerivaGem shows that the implied volatility of the call is 14.5%. For there to be no arbitrage, the put–call parity relationship in equation (20.1) must apply with q equal to the foreign risk-free rate. The price p of a European put option with a strike price of \$0.59 and maturity of 1 year therefore satisfies

$$p + 0.60e^{-0.10\times1} = 0.0236 + 0.59e^{-0.05\times1}$$

so that $p = 0.0419$. DerivaGem shows that, when the put has this price, its implied volatility is also 14.5%. This is what we expect from the analysis just given.

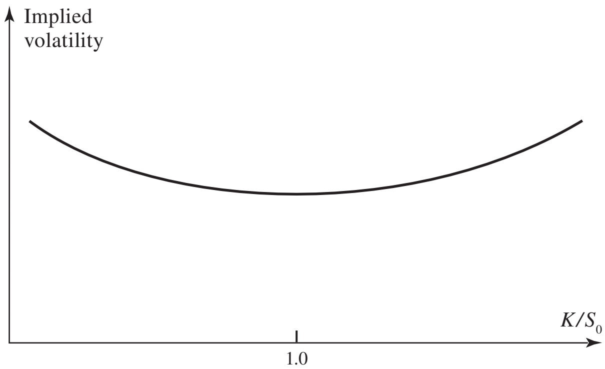
20.2 VOLATILITY SMILE FOR FOREIGN CURRENCY OPTIONS

The volatility smile used by traders to price foreign currency options tends to have the general form shown in Figure 20.1. The implied volatility is relatively low for at-the-money options. It becomes progressively higher as an option moves either into the money or out of the money.

In the appendix at the end of this chapter, we show how to determine the risk-neutral probability distribution for an asset price at a future time from the volatility smile given by options maturing at that time. We refer to this as the *implied distribution*. The volatility smile in Figure 20.1 corresponds to the implied distribution shown by the solid line in Figure 20.2. A lognormal distribution with the same mean and standard deviation as the implied distribution is shown by the dashed line in Figure 20.2. It can be seen that the implied distribution has heavier tails than the lognormal distribution.[1]

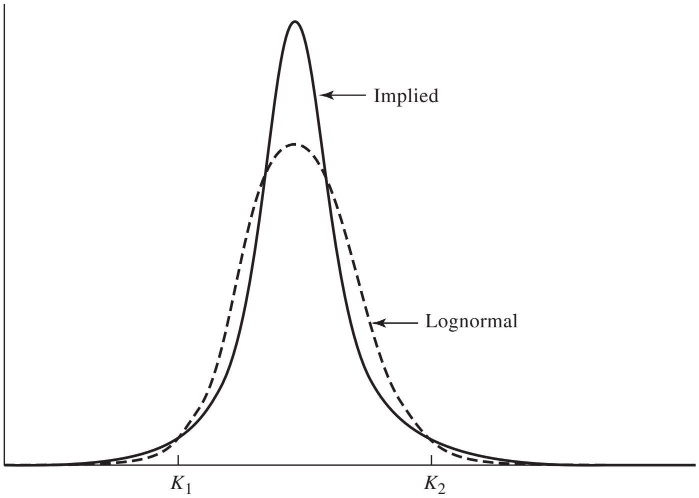
To see that Figures 20.1 and 20.2 are consistent with each other, consider first a deep-out-of-the-money call option with a high strike price of K_2 (K_2/S_0 well above 1.0). This option pays off only if the exchange rate proves to be above K_2. Figure 20.2 shows that the probability of this is higher for the implied probability distribution than for the lognormal distribution. We therefore expect the implied distribution to give a relatively high price for the option. A relatively high price leads to a relatively high implied volatility—and this is exactly what we observe in Figure 20.1 for the option. The two figures are therefore consistent with each other for high strike prices. Consider next a deep-out-of-the-money put option with a low strike price of K_1 (K_1/S_0 well below 1.0). This option pays off only if the exchange rate proves to be below K_1. Figure 20.2 shows that the probability of this is also higher for the implied probability distribution than for the lognormal distribution. We therefore expect the implied distribution to give a relatively high price, and a relatively high implied volatility, for this option as well. Again, this is exactly what we observe in Figure 20.1.

Figure 20.1 Volatility smile for foreign currency options (K = strike price, S_0 = current exchange rate).



[1] This is known as *kurtosis*. Note that, in addition to having a heavier tail, the implied distribution is more "peaked." Both small and large movements in the exchange rate are more likely than with the lognormal distribution. Intermediate movements are less likely.

Figure 20.2 Implied and lognormal distribution for foreign currency options.



Empirical Results

We have just shown that the volatility smile used by traders for foreign currency options implies that they consider that the lognormal distribution understates the probability of extreme movements in exchange rates. To test whether they are right, Table 20.1 examines the daily movements in 10 different exchange rates over a 10-year period between 2005 and 2015. The exchange rates are those between the U.S. dollar and the following currencies: Australian dollar, British pound, Canadian dollar, Danish krone, euro, Japanese yen, Mexican peso, New Zealand dollar, Swedish krona, and Swiss franc. The first step in the production of the table is to calculate the standard deviation of daily percentage change in each exchange rate. The next stage is to note how often the actual percentage change exceeded 1 standard deviation, 2 standard deviations, and so on. The final stage is to calculate how often this would have happened if the percentage changes had been normally distributed. (The lognormal model implies that percentage changes are almost exactly normally distributed over a one-day time period.)

Table 20.1 Percentage of days when daily exchange rate moves are greater than 1, 2, . . . , 6 standard deviations (SD = standard deviation of daily change).

	Real world	*Lognormal model*
>1 SD	23.32	31.73
>2 SD	4.67	4.55
>3 SD	1.30	0.27
>4 SD	0.49	0.01
>5 SD	0.24	0.00
>6 SD	0.13	0.00

Business Snapshot 20.1 Making Money from Foreign Currency Options

Black, Scholes, and Merton in their option pricing model assume that the underlying asset price has a lognormal distribution at future times. This is equivalent to the assumption that asset price changes over a short period of time, such as one day, are normally distributed. Suppose that most market participants are comfortable with the Black–Scholes–Merton assumptions for exchange rates. You have just done the analysis in Table 20.1 and know that the lognormal assumption is not a good one for exchange rates. What should you do?

The answer is that you should buy deep-out-of-the-money call and put options on a variety of different currencies and wait. These options will be relatively inexpensive and more of them will close in the money than the lognormal model predicts. The present value of your payoffs will on average be much greater than the cost of the options.

In the mid-1980s, a few traders knew about the heavy tails of foreign exchange probability distributions. Everyone else thought that the lognormal assumption of Black–Scholes–Merton was reasonable. The few traders who were well informed followed the strategy we have described—and made lots of money. By the late 1980s everyone realized that foreign currency options should be priced with a volatility smile and the trading opportunity disappeared.

Daily changes exceed 3 standard deviations on 1.30% of days. The lognormal model predicts that this should happen on only 0.27% of days. Daily changes exceed 4, 5, and 6 standard deviations on 0.49%, 0.24%, and 0.13% of days, respectively. The lognormal model predicts that we should hardly ever observe this happening. The table therefore provides evidence to support the existence of heavy tails (Figure 20.2) and the volatility smile used by traders (Figure 20.1). Business Snapshot 20.1 shows how you could have made money if you had done the analysis in Table 20.1 ahead of the rest of the market.

Reasons for the Smile in Foreign Currency Options

Why are exchange rates not lognormally distributed? Two of the conditions for an asset price to have a lognormal distribution are:

1. The volatility of the asset is constant.
2. The price of the asset changes smoothly with no jumps.

In practice, neither of these conditions is satisfied for an exchange rate. The volatility of an exchange rate is far from constant, and exchange rates frequently exhibit jumps, sometimes in response to the actions of central banks. It turns out that both a nonconstant volatility and jumps will have the effect of making extreme outcomes more likely.

The impact of jumps and nonconstant volatility depends on the option maturity. As the maturity of the option is increased, the percentage impact of a nonconstant volatility on prices becomes more pronounced, but its percentage impact on implied volatility usually becomes less pronounced. The percentage impact of jumps on both prices and the implied volatility becomes less pronounced as the maturity of the option is increased.[2] The result of all this is that the volatility smile becomes less pronounced as option maturity increases.

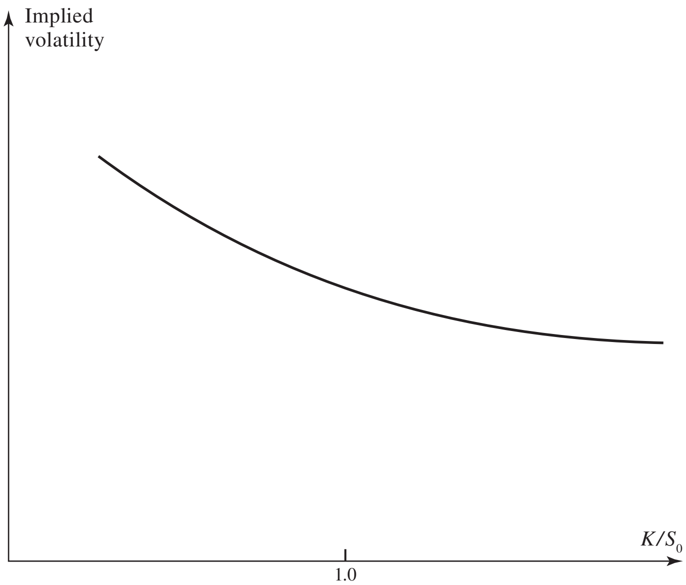
20.3 VOLATILITY SMILE FOR EQUITY OPTIONS

Prior to the crash of 1987, there was no marked volatility smile for equity options. Since 1987, the volatility smile used by traders to price equity options (both on individual stocks and on stock indices) has tended to look like that in Figure 20.3. This is sometimes referred to as a *volatility skew*. The volatility decreases as the strike price increases. The volatility used to price a low-strike-price option (i.e., a deep-out-of-the-money put or a deep-in-the-money call) is significantly higher than that used to price a high-strike-price option (i.e., a deep-in-the-money put or a deep-out-of-the-money call).

The volatility smile for equity options corresponds to the implied probability distribution given by the solid line in Figure 20.4. A lognormal distribution with the same mean and standard deviation as the implied distribution is shown by the dotted line. It can be seen that the implied distribution has a heavier left tail and a less heavy right tail than the lognormal distribution.

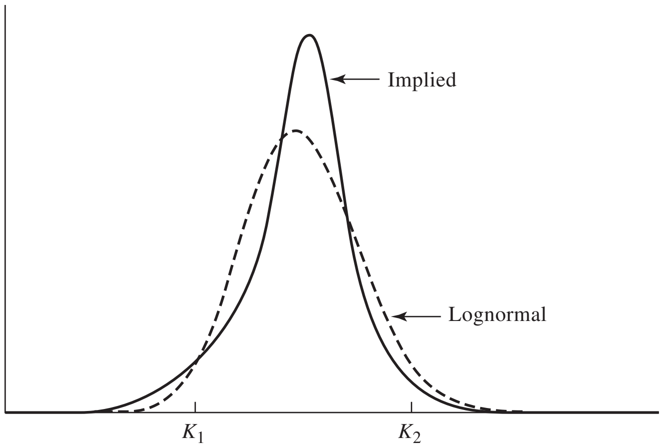
To see that Figures 20.3 and 20.4 are consistent with each other, we proceed as for Figures 20.1 and 20.2 and consider options that are deep out of the money. From Figure 20.4, a deep-out-of-the-money call with a strike price of K_2 (K_2/S_0 well above 1.0) has a lower price when the implied distribution is used than when the

Figure 20.3 Volatility smile for equities ($K =$ strike price, $S_0 =$ current equity price).



[2] When we look at sufficiently long-dated options, jumps tend to get "averaged out," so that the exchange rate distribution when there are jumps is almost indistinguishable from the one obtained when the exchange rate changes smoothly.

Figure 20.4 Implied distribution and lognormal distribution for equity options.



lognormal distribution is used. This is because the option pays off only if the stock price proves to be above K_2, and the probability of this is lower for the implied probability distribution than for the lognormal distribution. Therefore, we expect the implied distribution to give a relatively low price for the option. A relatively low price leads to a relatively low implied volatility—and this is exactly what we observe in Figure 20.3 for the option. Consider next a deep-out-of-the-money put option with a strike price of K_1. This option pays off only if the stock price proves to be below K_1 (K_1/S_0 well below 1.0). Figure 20.4 shows that the probability of this is higher for the implied probability distribution than for the lognormal distribution. We therefore expect the implied distribution to give a relatively high price, and a relatively high implied volatility, for this option. Again, this is exactly what we observe in Figure 20.3.

The Reason for the Smile in Equity Options

There is a negative correlation between equity prices and volatility.[3] As prices move down (up), volatilities tend to move up (down). There are several possible reasons for this. One concerns leverage. As equity prices move down (up), leverage increases (decreases) and as a result volatility increases (decreases). Another is referred to as the *volatility feedback effect*. As volatility increases (decreases) because of external factors, investors require a higher (lower) return and as a result the stock price declines (increases). A further explanation is crashophobia (see Business Snapshot 20.2).

Whatever the reason for the negative correlation, it means that stock price declines are accompanied by increases in volatility, making even greater declines possible. Stock price increases are accompanied by decreases in volatility, making further stock price increases less likely. This explains the heavy left tail and thin right tail of the implied distribution in Figure 20.4.

[3] For a machine learning investigation of this, see J. Cao, J. Chen, and J. Hull, "A Neural Network Approach to Understanding Implied Volatility Movements," *Quantitative Finance*, 20, 9 (2020): 1405–13.

Business Snapshot 20.2 Crashophobia

It is interesting that the pattern in Figure 20.3 for equities has existed only since the stock market crash of October 1987. Prior to October 1987, implied volatilities were much less dependent on strike price. This has led Mark Rubinstein to suggest that one reason for the equity volatility smile may be "crashophobia." Traders are concerned about the possibility of another crash similar to October 1987, and they price options accordingly.

There is some empirical support for this explanation. Declines in the S&P 500 tend to be accompanied by a steepening of the volatility skew. When the S&P increases, the skew tends to become less steep.

20.4 ALTERNATIVE WAYS OF CHARACTERIZING THE VOLATILITY SMILE

There are a number of ways of characterizing the volatility smile. Sometimes it is shown as the relationship between implied volatility and strike price K. However, this relationship depends on the price of the asset. As the price of the asset increases (decreases), the central at-the-money strike price increases (decreases) so that the curve relating the implied volatility to the strike price moves to right (left).[4] For this reason the implied volatility is often plotted as a function of the strike price divided by the current asset price, K/S_0. This is what we have done Figures 20.1 and 20.3.

A refinement of this is to calculate the volatility smile as the relationship between the implied volatility and K/F_0, where F_0 is the forward price of the asset for a contract maturing at the same time as the options that are considered. Traders also often define an "at-the-money" option as an option where $K = F_0$, not as an option where $K = S_0$. The argument for this is that F_0, not S_0, is the expected stock price on the option's maturity date in a risk-neutral world.

Yet another approach to defining the volatility smile is as the relationship between the implied volatility and the delta of the option (where delta is defined as in Chapter 19). This approach sometimes makes it possible to apply volatility smiles to options other than European and American calls and puts. When the approach is used, an at-the-money option is then defined as a call option with a delta of 0.5 or a put option with a delta of -0.5. These are referred to as "50-delta options."

20.5 THE VOLATILITY TERM STRUCTURE AND VOLATILITY SURFACES

Traders allow the implied volatility to depend on time to maturity as well as strike price. Implied volatility tends to be an increasing function of maturity when short-dated volatilities are historically low. This is because there is then an expectation that volatilities will increase. Similarly, volatility tends to be a decreasing function of

[4] Research by Derman suggests that this adjustment is sometimes "sticky" in the case of exchange-traded options. See E. Derman, "Regimes of Volatility," *Risk*, April 1999: 55–59.

Table 20.2 Volatility surface.

	K/S_0				
	0.90	*0.95*	*1.00*	*1.05*	*1.10*
1 month	14.2	13.0	12.0	13.1	14.5
3 month	14.0	13.0	12.0	13.1	14.2
6 month	14.1	13.3	12.5	13.4	14.3
1 year	14.7	14.0	13.5	14.0	14.8
2 year	15.0	14.4	14.0	14.5	15.1
5 year	14.8	14.6	14.4	14.7	15.0

maturity when short-dated volatilities are historically high. This is because there is then an expectation that volatilities will decrease.

Volatility surfaces combine volatility smiles with the volatility term structure to tabulate the volatilities appropriate for pricing an option with any strike price and any maturity. An example of a volatility surface that might be used for foreign currency options is given in Table 20.2.

One dimension of Table 20.2 is K/S_0; the other is time to maturity. The main body of the table shows implied volatilities calculated from the Black–Scholes–Merton model. At any given time, some of the entries in the table are likely to correspond to options for which reliable market data are available. The implied volatilities for these options are calculated directly from their market prices and entered into the table. The rest of the table is typically determined using interpolation. The table shows that the volatility smile becomes less pronounced as the option maturity increases. As mentioned earlier, this is what is observed for currency options. (It is also what is observed for options on most other assets.)

When a new option has to be valued, financial engineers look up the appropriate volatility in the table. For example, when valuing a 9-month option with a K/S_0 ratio of 1.05, a financial engineer would interpolate between 13.4 and 14.0 in Table 20.2 to obtain a volatility of 13.7%. This is the volatility that would be used in the Black–Scholes–Merton formula or a binomial tree. When valuing a 1.5-year option with a K/S_0 ratio of 0.925, a two-dimensional (bilinear) interpolation would be used to give an implied volatility of 14.525%.

The shape of the volatility smile depends on the option maturity. As illustrated in Table 20.2, the smile tends to become less pronounced as the option maturity increases. Define T as the time to maturity and F_0 as the forward price of the asset for a contract maturing at the same time as the option. Some financial engineers choose to define the volatility smile as the relationship between implied volatility and

$$\frac{1}{\sqrt{T}} \ln\left(\frac{K}{F_0}\right)$$

rather than as the relationship between the implied volatility and K. The smile is then usually much less dependent on the time to maturity.

20.6 MINIMUM VARIANCE DELTA

The formulas for delta and other Greek letters in Chapter 19 assume that the implied volatility remains the same when the asset price changes. This is not what is usually expected to happen. Consider, for example, a stock or stock index option.

As explained in Section 20.3, there is a negative correlation between equity prices and volatility. The delta that takes this relationship between implied volatilities and equity prices into account is referred to as the *minimum variance delta*. It is:

$$\Delta_{\mathrm{MV}} = \frac{\partial f_{\mathrm{BSM}}}{\partial S} + \frac{\partial f_{\mathrm{BSM}}}{\partial \sigma_{\mathrm{imp}}} \frac{\partial E(\sigma_{\mathrm{imp}})}{\partial S}$$

where f_{BSM} is the Black–Scholes–Merton price of the option, σ_{imp} is the option's implied volatility, $E(\sigma_{\mathrm{imp}})$ denotes the expectation of σ_{imp} as a function of the equity price, S. This gives

$$\Delta_{\mathrm{MV}} = \Delta_{\mathrm{BSM}} + \mathcal{V}_{\mathrm{BSM}} \frac{\partial E(\sigma_{\mathrm{imp}})}{\partial S}$$

where Δ_{BSM} and $\mathcal{V}_{\mathrm{BSM}}$ are the delta and vega calculated from the Black–Scholes–Merton (constant volatility) model. Because $\mathcal{V}_{\mathrm{BSM}}$ is positive and, as we have just explained $\partial E(\sigma_{\mathrm{imp}})/\partial S$ is negative, the minimum variance delta is less than the Black–Scholes–Merton delta.[5]

20.7 THE ROLE OF THE MODEL

How important is the option-pricing model if traders are prepared to use a different volatility for every option? It can be argued that the Black–Scholes–Merton model is no more than a sophisticated interpolation tool used by traders for ensuring that an option is priced consistently with the market prices of other actively traded options. If traders stopped using Black–Scholes–Merton and switched to another plausible model, then the volatility surface and the shape of the smile would change, but arguably the dollar prices quoted in the market would not change appreciably. Greek letters and therefore hedging strategies do depend on the model used. An unrealistic model is liable to lead to poor hedging.

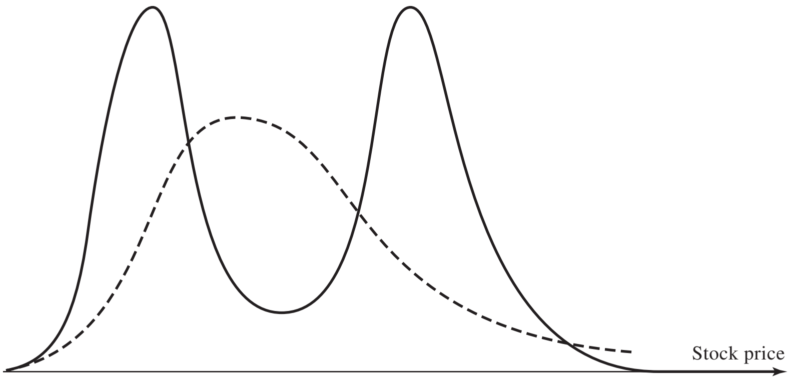
Models have most effect on the pricing of derivatives when similar derivatives do not trade actively in the market. For example, the pricing of many of the nonstandard exotic derivatives we will discuss in later chapters is model-dependent.

20.8 WHEN A SINGLE LARGE JUMP IS ANTICIPATED

Let us now consider an example of how an unusual volatility smile might arise in equity markets. Suppose that a stock price is currently \$50 and an important news announcement due in a few days is expected either to increase the stock price by \$8 or to reduce it by \$8. (This announcement could concern the outcome of a takeover attempt or the

[5] For a further discussion of this, see, for example, J. C. Hull and A. White, "Optimal Delta Hedging of Options," *Journal of Banking and Finance*, 82 (September 2017), 180–190.

Figure 20.5 Effect of a single large jump. The solid line is the true distribution; the dashed line is the lognormal distribution.



verdict in an important lawsuit.) The probability distribution of the stock price in, say, 1 month might then consist of a mixture of two lognormal distributions, the first corresponding to favorable news, the second to unfavorable news. The situation is illustrated in Figure 20.5. The solid line shows the mixture-of-lognormals distribution for the stock price in 1 month; the dashed line shows a lognormal distribution with the same mean and standard deviation as this distribution.

The true probability distribution is bimodal (certainly not lognormal). One easy way to investigate the general effect of a bimodal stock price distribution is to consider the extreme case where there are only two possible future stock prices. This is what we will now do.

Suppose that the stock price is currently \$50 and it is known that in 1 month it will be either \$42 or \$58. Suppose further that the risk-free rate is 12% per annum. The situation is illustrated in Figure 20.6. Options can be valued using the binomial model from Chapter 13. In this case, $u = 1.16$, $d = 0.84$, $a = 1.0101$, and $p = 0.5314$. The results from valuing a range of different options are shown in Table 20.3. The first column shows alternative strike prices; the second shows prices of 1-month European call options; the third shows the prices of one-month European put option prices; and the fourth shows

Figure 20.6 Change in stock price in 1 month.

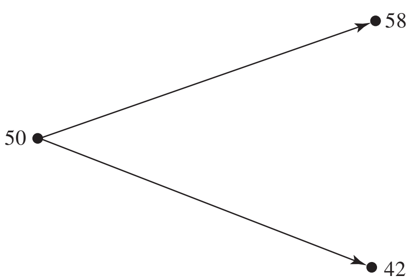


Table 20.3 Implied volatilities in situation where it is known that the stock price will move from \$50 to either \$42 or \$58.

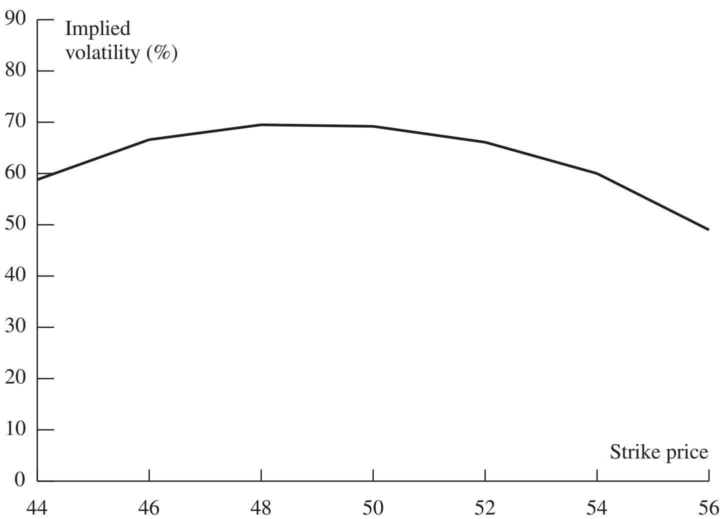
Strike price (\$)	*Call price (\$)*	*Put price (\$)*	*Implied volatility (%)*
42	8.42	0.00	0.0
44	7.37	0.93	58.8
46	6.31	1.86	66.6
48	5.26	2.78	69.5
50	4.21	3.71	69.2
52	3.16	4.64	66.1
54	2.10	5.57	60.0
56	1.05	6.50	49.0
58	0.00	7.42	0.0

implied volatilities. (As shown in Section 20.1, the implied volatility of a European put option is the same as that of a European call option when they have the same strike price and maturity.) Figure 20.7 displays the volatility smile from Table 20.3. It is actually a "frown" (the opposite of that observed for currencies) with volatilities declining as we move out of or into the money. The volatility implied from an option with a strike price of 50 will overprice an option with a strike price of 44 or 56.

SUMMARY

The Black–Scholes–Merton model and its extensions assume that the probability distribution of the underlying asset at any given future time is lognormal. This

Figure 20.7 Volatility smile for situation in Table 20.3.



assumption is not the one made by traders. They assume the probability distribution of an equity price has a heavier left tail and a less heavy right tail than the lognormal distribution. They also assume that the probability distribution of an exchange rate has a heavier right tail and a heavier left tail than the lognormal distribution.

Traders use volatility smiles to allow for nonlognormality. The volatility smile defines the relationship between the implied volatility of an option and its strike price. For equity options, the volatility smile tends to be downward sloping. This means that out-of-the-money puts and in-the-money calls tend to have high implied volatilities whereas out-of-the-money calls and in-the-money puts tend to have low implied volatilities. For foreign currency options, the volatility smile is U-shaped. Both out-of-the-money and in-the-money options have higher implied volatilities than at-the-money options.

Often traders also use a volatility term structure. The implied volatility of an option then depends on its life. When volatility smiles and volatility term structures are combined, they produce a volatility surface. This defines implied volatility as a function of both the strike price and the time to maturity.

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Practice Questions

20.1. What volatility smile is likely to be observed when:

(a) Both tails of the stock price distribution are less heavy than those of the lognormal distribution?

(b) The right tail is heavier, and the left tail is less heavy, than that of a lognormal distribution?

20.2. What volatility smile is likely to be caused by jumps in the underlying asset price? Is the pattern likely to be more pronounced for a 2-year option than for a 3-month option?

20.3. A European call and put option have the same strike price and time to maturity. The call has an implied volatility of 30% and the put has an implied volatility of 25%. What trades would you do?

20.4. Explain carefully why a distribution with a heavier left tail and less heavy right tail than the lognormal distribution gives rise to a downward sloping volatility smile.

20.5. The market price of a European call is \$3.00 and its price given by Black–Scholes–Merton model with a volatility of 30% is \$3.50. The price given by this Black–Scholes–Merton model for a European put option with the same strike price and time to maturity is \$1.00. What should the market price of the put option be? Explain the reasons for your answer.

20.6. Explain what is meant by "crashophobia."

20.7. A stock price is currently \$20. Tomorrow, news is expected to be announced that will either increase the price by \$5 or decrease the price by \$5. What are the problems in using Black–Scholes–Merton to value 1-month options on the stock?

20.8. What volatility smile is likely to be observed for 6-month options when the volatility is uncertain and positively correlated with the stock price?

20.9. Explain the problems in testing a stock option pricing model empirically.

20.10. Suppose that a central bank's policy is to allow an exchange rate to fluctuate between 0.97 and 1.03. What pattern of implied volatilities for options on the exchange rate would you expect to see?

20.11. Option traders sometimes refer to deep-out-of-the-money options as being options on volatility. Why do you think they do this?

20.12. A European call option on a certain stock has a strike price of \$30, a time to maturity of 1 year, and an implied volatility of 30%. A European put option on the same stock has a strike price of \$30, a time to maturity of 1 year, and an implied volatility of 33%. What is the arbitrage opportunity open to a trader? Does the arbitrage work only when the lognormal assumption underlying Black–Scholes–Merton holds? Explain carefully the reasons for your answer.

20.13. Suppose that the result of a major lawsuit affecting a company is due to be announced tomorrow. The company's stock price is currently \$60. If the ruling is favorable to the company, the stock price is expected to jump to \$75. If it is unfavorable, the stock is expected to jump to \$50. What is the risk-neutral probability of a favorable ruling?

Assume that the volatility of the company's stock will be 25% for 6 months after the ruling if the ruling is favorable and 40% if it is unfavorable. Use DerivaGem to calculate the relationship between implied volatility and strike price for 6-month European options on the company today. The company does not pay dividends. Assume that the 6-month risk-free rate is 6%. Consider call options with strike prices of $30, $40, $50, $60, $70, and $80.

20.14. An exchange rate is currently 0.8000. The volatility of the exchange rate is quoted as 12% and interest rates in the two countries are the same. Using the lognormal assumption, estimate the probability that the exchange rate in 3 months will be (a) less than 0.7000, (b) between 0.7000 and 0.7500, (c) between 0.7500 and 0.8000, (d) between 0.8000 and 0.8500, (e) between 0.8500 and 0.9000, and (f) greater than 0.9000. Based on the volatility smile usually observed in the market for exchange rates, which of these estimates would you expect to be too low and which would you expect to be too high?

20.15. A stock price is $40. A 6-month European call option on the stock with a strike price of $30 has an implied volatility of 35%. A 6-month European call option on the stock with a strike price of $50 has an implied volatility of 28%. The 6-month risk-free rate is 5% and no dividends are expected. Explain why the two implied volatilities are different. Use DerivaGem to calculate the prices of the two options. Use put–call parity to calculate the prices of 6-month European put options with strike prices of $30 and $50. Use DerivaGem to calculate the implied volatilities of these two put options.

20.16. "The Black–Scholes–Merton model is used by traders as an interpolation tool." Discuss this view.

20.17. Using Table 20.2, calculate the implied volatility a trader would use for an 8-month option with $K/S_0 = 1.04$.

20.18. A company's stock is selling for $4. The company has no outstanding debt. Analysts consider the liquidation value of the company to be at least $300,000 and there are 100,000 shares outstanding. What volatility smile would you expect to see?

20.19. Data for a number of foreign currencies are provided on the author's website:

http://www-2.rotman.utoronto.ca/~hull/data

Choose a currency and use the data to produce a table similar to Table 20.1.

20.20. Data for a number of stock indices are provided on the author's website:

http://www-2.rotman.utoronto.ca/~hull/data

Choose an index and test whether a three-standard-deviation down movement happens more often than a three-standard-deviation up movement.

20.21. Consider a European call and a European put with the same strike price and time to maturity. Show that they change in value by the same amount when the volatility increases from a level σ_1 to a new level σ_2 within a short period of time. (*Hint*: Use put–call parity.)

20.22. An exchange rate is currently 1.0 and the implied volatilities of 6-month European options with strike prices 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3 are 13%, 12%, 11%, 10%, 11%, 12%, 13%. The domestic and foreign risk-free rates are both 2.5%. Calculate the implied probability distribution using an approach similar to that used for Example 20A.1 in the appendix to this chapter. Compare it with the implied distribution where all the implied volatilities are 11.5%.

20.23. Using Table 20.2, calculate the implied volatility a trader would use for an 11-month option with $K/S_0 = 0.98$.

APPENDIX

DETERMINING IMPLIED RISK-NEUTRAL DISTRIBUTIONS FROM VOLATILITY SMILES

The price of a European call option on an asset with strike price K and maturity T is given by

$$c = e^{-rT} \int_{S_T=K}^{\infty} (S_T - K)\, g(S_T) dS_T$$

where r is the interest rate (assumed constant), S_T is the asset price at time T, and g is the risk-neutral probability density function of S_T. Differentiating once with respect to K gives

$$\frac{\partial c}{\partial K} = -e^{-rT} \int_{S_T=K}^{\infty} g(S_T) dS_T$$

Differentiating again with respect to K gives

$$\frac{\partial^2 c}{\partial K^2} = e^{-rT}\, g(K)$$

This shows that the probability density function g is given by

$$g(K) = e^{rT} \frac{\partial^2 c}{\partial K^2} \qquad \textbf{(20A.1)}$$

This result, which is from Breeden and Litzenberger (1978), allows risk-neutral probability distributions to be estimated from volatility smiles.[6] Suppose that c_1, c_2, and c_3 are the prices of T-year European call options with strike prices of $K - \delta$, K, and $K + \delta$, respectively. Assuming δ is small, an estimate of $g(K)$, obtained by approximating the partial derivative in equation (20A.1), is

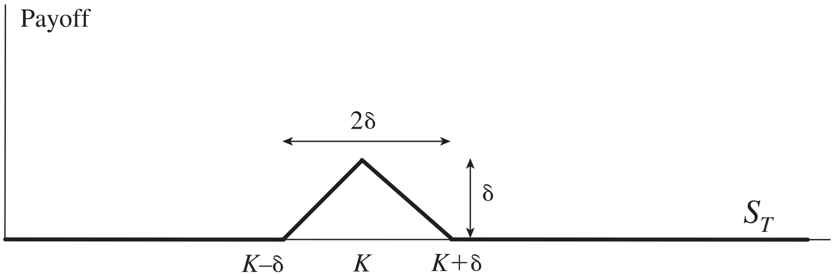
$$e^{rT} \frac{c_1 + c_3 - 2c_2}{\delta^2}$$

For another way of understanding this formula, suppose you set up a butterfly spread with strike prices $K - \delta$, K, and $K + \delta$, and maturity T. This means that you buy a call with strike price $K - \delta$, buy a call with strike price $K + \delta$, and sell two calls with strike price K. The value of your position is $c_1 + c_3 - 2c_2$. The value of the position can also be calculated by integrating the payoff over the risk-neutral probability distribution, $g(S_T)$, and discounting at the risk-free rate. The payoff is shown in Figure 20A.1. Since δ is small, we can assume that $g(S_T) = g(K)$ in the whole of the range $K - \delta < S_T < K + \delta$, where the payoff is nonzero. The area under the "spike" in Figure 20A.1 is $0.5 \times 2\delta \times \delta = \delta^2$. The value of the payoff (when δ is small) is therefore $e^{-rT} g(K)\delta^2$. It follows that

$$e^{-rT} g(K)\delta^2 = c_1 + c_3 - 2c_2$$

[6] See D. T. Breeden and R. H. Litzenberger, "Prices of State-Contingent Claims Implicit in Option Prices," *Jounal of Business*, 51 (1978), 621–51.

Figure 20A.1 Payoff from butterfly spread.



which leads directly to

$$g(K) = e^{rT}\frac{c_1 + c_3 - 2c_2}{\delta^2} \tag{20A.2}$$

Example 20A.1

Suppose that the price of a non-dividend-paying stock is \$10, the risk-free interest rate is 3%, and the implied volatilities of 3-month European options with strike prices of \$6, \$7, \$8, \$9, \$10, \$11, \$12, \$13, \$14 are 30%, 29%, 28%, 27%, 26%, 25%, 24%, 23%, 22%, respectively. One way of applying the above results is as follows. Assume that $g(S_T)$ is constant between $S_T = 6$ and $S_T = 7$, constant between $S_T = 7$ and $S_T = 8$, and so on. Define:

$$\begin{aligned}
g(S_T) &= g_1 \quad \text{for} \quad 6 \le S_T < 7 \\
g(S_T) &= g_2 \quad \text{for} \quad 7 \le S_T < 8 \\
g(S_T) &= g_3 \quad \text{for} \quad 8 \le S_T < 9 \\
g(S_T) &= g_4 \quad \text{for} \quad 9 \le S_T < 10 \\
g(S_T) &= g_5 \quad \text{for} \quad 10 \le S_T < 11 \\
g(S_T) &= g_6 \quad \text{for} \quad 11 \le S_T < 12 \\
g(S_T) &= g_7 \quad \text{for} \quad 12 \le S_T < 13 \\
g(S_T) &= g_8 \quad \text{for} \quad 13 \le S_T < 14
\end{aligned}$$

The value of g_1 can be calculated by interpolating to get the implied volatility for a 3-month option with a strike price of \$6.5 as 29.5%. This means that options with strike prices of \$6, \$6.5, and \$7 have implied volatilities of 30%, 29.5%, and 29%, respectively. From DerivaGem their prices are \$4.045, \$3.549, and \$3.055, respectively. Using equation (20A.2), with $K = 6.5$ and $\delta = 0.5$, gives

$$g_1 = \frac{e^{0.03\times 0.25}(4.045 + 3.055 - 2 \times 3.549)}{0.5^2} = 0.0057$$

Similar calculations show that

$$g_2 = 0.0444, \quad g_3 = 0.1545, \quad g_4 = 0.2781$$

$$g_5 = 0.2813, \quad g_6 = 0.1659, \quad g_7 = 0.0573, \quad g_8 = 0.0113$$

Figure 20A.2 Implied probability distribution for Example 20A.1.

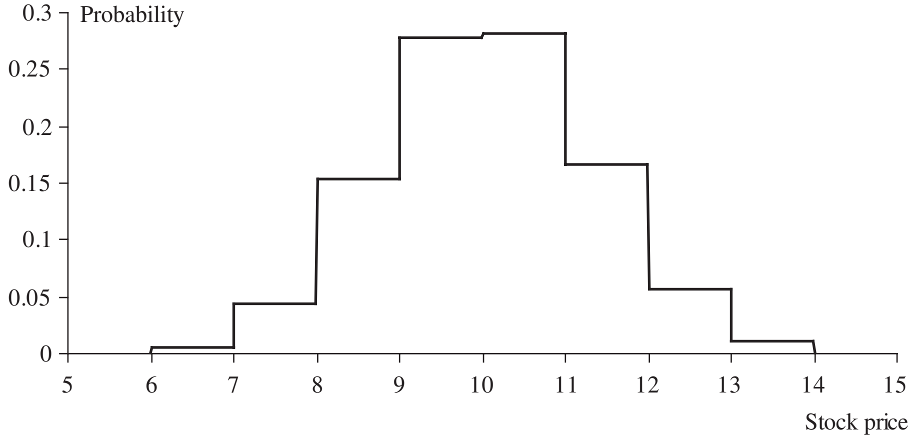


Figure 20A.2 displays the implied distribution. (Note that the area under the probability distribution is 0.9985. The probability that $S_T < 6$ or $S_T > 14$ is therefore 0.0015.) Although not obvious from Figure 20A.2, the implied distribution does have a heavier left tail and less heavy right tail than a lognormal distribution. For the lognormal distribution based on a single volatility of 26%, the probability of a stock price between \$6 and \$7 is 0.0031 (compared with 0.0057 in Figure 20A.2) and the probability of a stock price between \$13 and \$14 is 0.0167 (compared with 0.0113 in Figure 20A.2).